INTRODUCTION

Learning Outcome

- describe a simple linear regression model, how the least squares criterion is used to estimate regression coefficients, and the interpretation of these coefficients

Financial analysts often need to examine whether a variable is useful for explaining another variable. For example, the analyst may want to know whether earnings or cash flow growth help explain a company's market value. **Regression analysis** is a tool for examining this type of issue.

Linear regression allows us to test hypotheses about the relationship between two variables by quantifying the strength of the relationship between the two variables, and to use one variable to make predictions about the other variable. Our focus is on linear regression with a single independent variable—that is, simple linear regression.

Learning Module Overview

- Simple linear regression is a mathematical process for determining how the variation in one variable can explain the variation in another variable.
- The variable we wish to explain is called the dependent variable, and the variable that we use to explain the dependent variable is called the independent variable.
- Simple linear regression uses the ordinary least squares approach to calculate the slope and intercept parameters that characterize a linear relationship between the two variables.
- Simple linear regression requires that we make four assumptions: linearity, homoskedasticity, independence, and normality.
- Linearity requires that the regression residuals be random and that the independent variable not be random. Homoskedasticity, which refers to variance being constant across observations, cannot be assumed when we see residuals clustering in multiple groups because the clustering indicates multiple regimes with different variances within our time period.
- Independence means that the X-Y pairs are uncorrelated; a pattern in a plot of the residuals (e.g., seasonality) suggests that there is autocorrelation across observations and that we cannot assume independence. Normality means that residuals must be normally distributed and does not require that the data itself be normally distributed. Non-normality is of particular concern for small sample sizes, but for large sample sizes, the central limit theorem tells us that we may be able to relax the normality requirement.
- The total variation in the dependent variable, called the sum of squares total (SST), can be decomposed into two parts: the explained variation, called the sum of squares regression (SSR), and the unexplained variation, called the sum of squares error (SSE).
- There are several ways to measure a regression model's goodness of fit. These include the coefficient of determination, the *F*-statistic for the test of fit, and the standard error of the regression.
- Hypothesis testing can be used to determine, at a specified confidence level, whether the slope or intercept differs from zero or another specified value, or whether the slope is positive or negative. We can use indicator variables to determine whether our regression parameters differ between data points that either have or do not have a particular characteristic (e.g., monthly price data in cases in which only some months have earnings announcements).
- An analysis of variance (ANOVA) table presents the sums of squares, degrees of freedom, mean squares, and *F*-statistic for a regression model.
- The standard error of the estimate is a measure of the distance between the observed values of the dependent variable and those predicted from the estimated regression. The smaller this value, the better the fit of the model.
- The standard error of the forecast is used to provide an interval estimate around the estimated regression line. It is necessary because the regression line does not describe the relationship between the dependent and independent variables perfectly.
- The simple linear regression model on non-linear data can be adjusted by using different functional forms that transform the dependent or independent variables.
- Three common functional forms for transforming data include the log-lin model, the lin-log model, and the log-log model.
- The key to fitting the appropriate functional form of a simple linear regression is examining the goodness-of-fit measures—the coefficient of determination (R^2), the *F*-statistic, and the standard error of the estimate (s_e)—as well as examining whether there are patterns in the residuals.

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